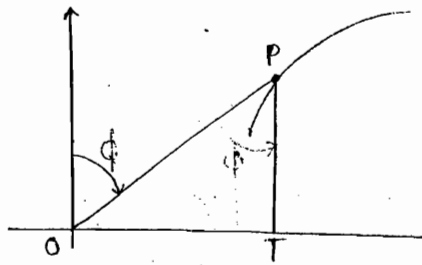


$$T_{sh} = \frac{60}{2\pi N} \cdot \ell(p\phi) \times \text{power scale} \cdot N-m$$

Power Factor $\rightarrow$



From ΔOPT

$$\cos\phi = \frac{PT}{OP}$$

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* starting methods $\rightarrow$ When rated vol. is applied across the stator of IM during starting $N=0, S=1$ it is eq. to applying rated vol. across Xmer 1^o when its 2^o is SC.

The rotor will have huge current which comes through stator.

Therefore the motor draws extremely large current from supply which is not acceptable because it produces vol. dip in supply lines.

* It may not damage the motor as they are rugged in construction & accelerate quickly & the starting current is inrush.

(1.) Direct online starting (DOL)

(2.) Stator resistance/Reactance starting

(3.) Auto Xmers.

(4.) Star-delta

(5.) Rotor resistance

} Reduced vol. starting

(1.) Direct online starting $\rightarrow$

* popular for 1ϕ & upto 5kw For 3ϕ because of cost is less.

* Push on/ push off

Overload $\rightarrow$ Bimetallic

⊗ Underload $\rightarrow$ When power is not present

* It is not reduced vol. starting.

* It has a push on/ push off switch where the stator is directly connected to

supply.

* starting current is very high but it is inrush.

As the motor starts quickly.

* It is only suitable to small rating.

most popular under 2kw segment (1 ϕ) as it is low cost.

* There are 2 protection:-

(1) Overload protection using thermal elements like bimetallic.

(2) Undervolt protection to prevent the motor from unintentional start after a power failure.

(2) Stator resistance / reactors starting $\rightarrow$

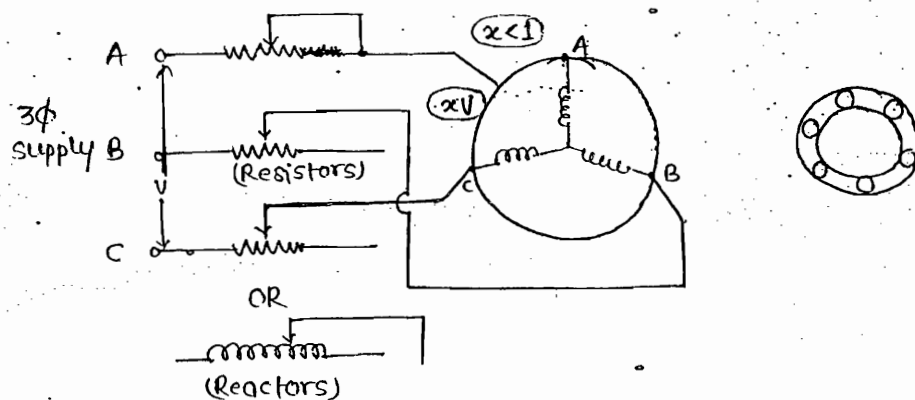
When we supply the stator excessive amt. of current flows in the rotor.

So;

$$I_2 = \frac{E_2}{R_2 + jX_2} \rightarrow \text{Fixed}$$

$$E_2 \downarrow, I_2 \downarrow$$

$$\text{So, } E_2 \downarrow \rightarrow \phi \downarrow \rightarrow V_1 \downarrow$$



$$T_g = \frac{60}{2\pi N_s} \frac{\text{Rot. cu loss}}{s}$$

$$T_g = \frac{60}{2\pi N_s} \frac{3(I_2^2 R_2)}{s}$$

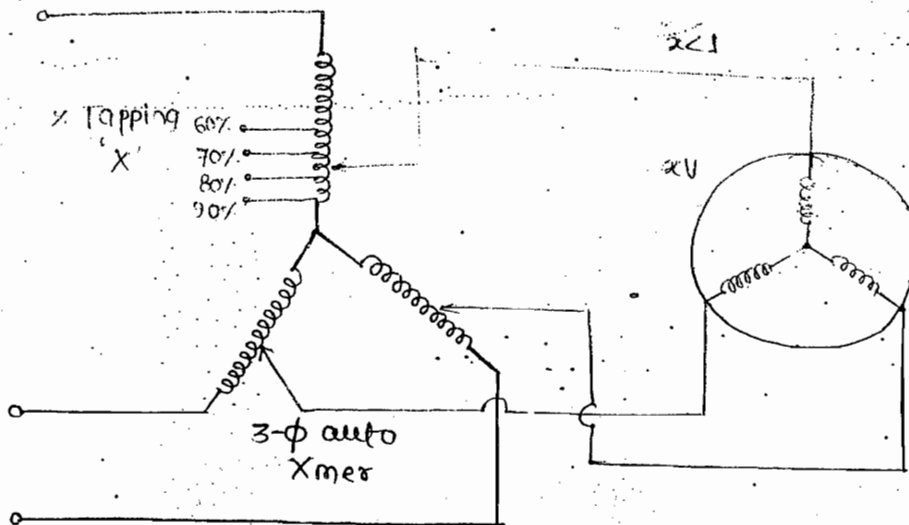
$$T_{st} \propto \frac{I_{st}^2}{s_{st}} \quad T_f \propto \frac{I_f^2}{s_f} \quad (s_{st}=1)$$

$$\frac{T_{st}}{T_f} = \left(\frac{I_{st}}{I_f} \right)^2 \cdot s_f$$

$$\frac{T_{st}}{T_f} = x^2 \left(\frac{I_{st}}{I_f} \right)^2 \cdot s_f$$

- * By adding some external resistance in series with stator some vol. gets dropped & reduced vol. will appear across stator which correspondingly control the starting current.
- * Reactors also can be used where η is not greatly effected but pf of the motor will be low lagging & starting torque also less.
- * The effect of starting torque is expressed in above relation.

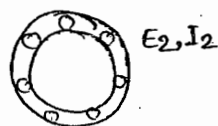
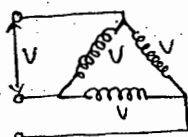
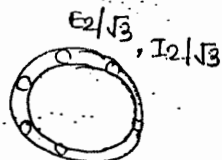
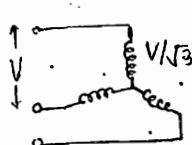
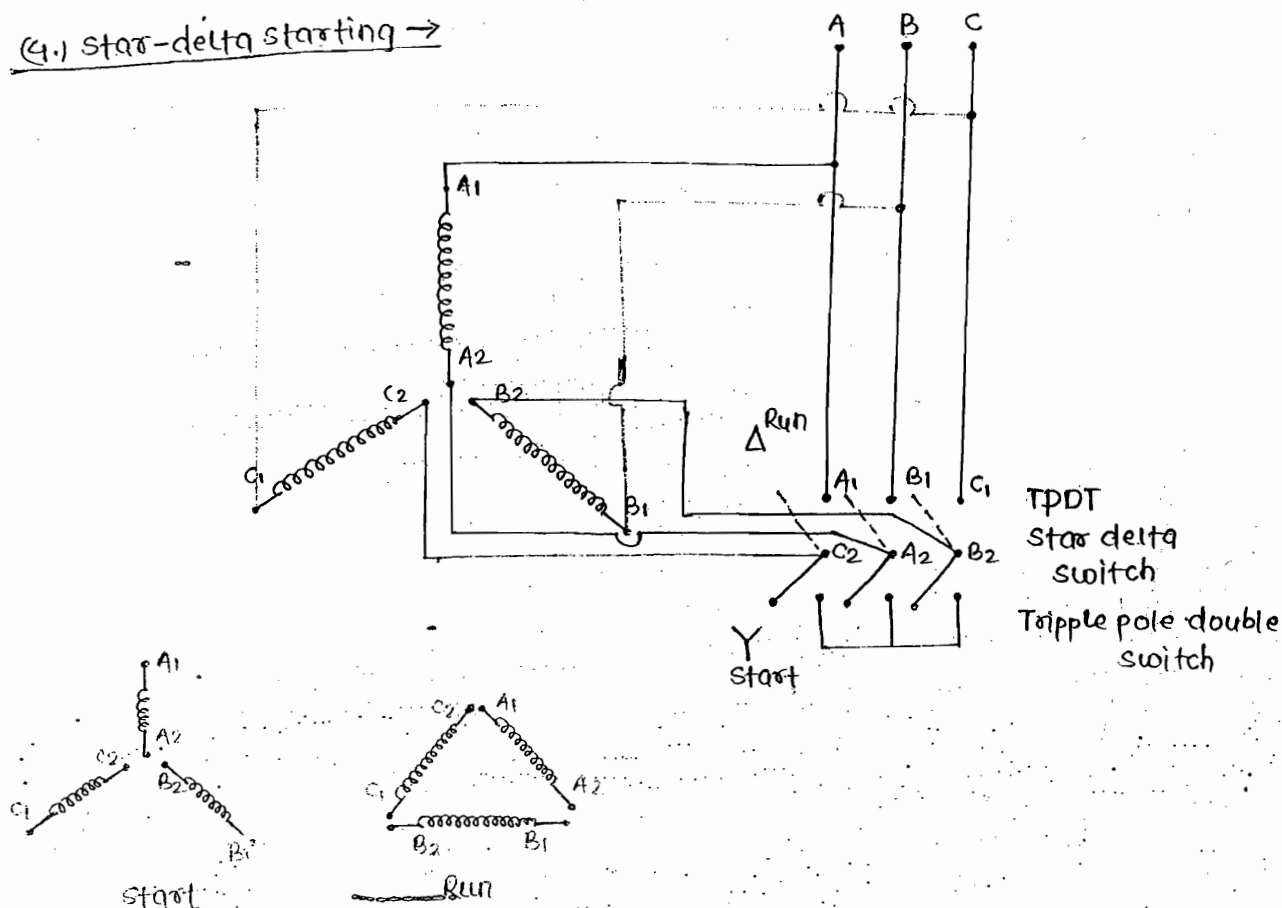
(3) Auto Xmer starting $\rightarrow$



- * This is generally preferred for starting large rating IMs due to its flexibility & high η but expensive method.
- * It requires a suitable rating 3 ϕ auto Xmers.
- * Depending on % tapping ($x < 1$)
depending on tapping starting current control & effect of starting torque expressed

$$\frac{T_{st}}{T_f} = x^2 \left(\frac{I_{st}}{I_f} \right)^2 \cdot s_f$$

(4.) star-delta starting →



* It gives one step control on the starting current.

It is most simple & economic starter with a TPDT switch only.

The stator 3 ϕ wdg is connected as shown.

It is suitable for motors which are designed for Δ under running condⁿ.
1st the stator is connected the λ which results in reduced vol. & current in rotor about 57.7% only ($V_{ph} = V/\sqrt{3}$).

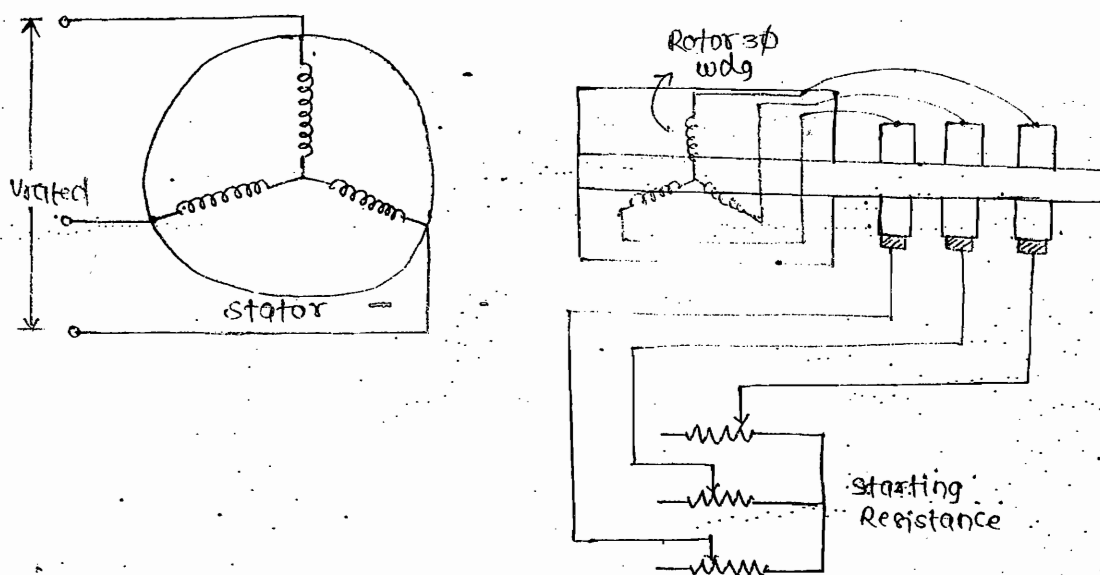
Under running the switch should be thrown out to Δ atleast the motor reach 90% of rated speed for normal rated operating condⁿ ($V_L = V_{ph}$)

* Not suitable for extra high vol. rating greater than 3.3kV because those motors stators are preferably designed for Δ .

$$\frac{T_{st}}{T_f} = \frac{1}{3} \left(\frac{I_{st}}{I_f} \right)^2 \cdot s_f$$

* The ratio of starting to fl torque is expressed above.

5.1 Rotor resistance starting $\rightarrow$



$$T \propto \underset{\substack{\uparrow \text{very} \\ \downarrow \text{Rated}}}{E_2} I_2 \cos \phi_2 \quad (\text{Because of resistance})$$

* This is technically best way to start IM.

* It is fundamental method of starting slip ring motors by inserting some resistance into rotor of suitable value through slip rings & brushes.

* This resistance will control the rotor current & motor current during starting. While reducing the starting current to a required value starting torque is increased because the improved PF will dominate.

* However rated vol. is applied across the stator.

This are preferred for high starting torque appln like electric traction

(52/32) (c)

$$\frac{T_{st}}{T_F} = x^2 \left(\frac{I_{st}}{I_F} \right)^2 S_F$$

$$1.5 = x^2 (7)^2 (0.05)$$

$$x = 78.25\%$$

(53/32)

(b.) $\frac{T_{st}}{T_F} = \frac{1}{3} \left(\frac{I_{st}}{I_F} \right)^2 S_F$

$$= \frac{1}{3} (7)^2 (0.05)$$

$$= 0.816$$

(54/32)

(c) Because of no given any starting method, use basic formula.

$$\left(\frac{I_{st}}{I_F} \right) = \left(\frac{I_{st}}{I_F} \right)^2 S_F$$

$$0.5 = \left(\frac{I_{st}}{I_F} \right)^2 \times 0.05$$

$$\left(\frac{I_{st}}{I_F} \right) = 3.16$$

(60/33)

(d.)

$$\frac{T_{st}}{T_F} = \left(\frac{I_{st}}{I_F} \right)^2 S_F$$

$$1 = \left(\frac{I_{st}}{I_F} \right)^2 (0.04)$$

$$\left(\frac{I_{st}}{I_F} \right) = 5$$

(64/33)

(d.)

$$(0.3)^2 \rightarrow 80 \text{ N-m}$$

$$(0.6)^2 \rightarrow ?$$

* Speed control →

* These are not popular like dc shunt motors when compared to effective speed control. even though the T-S c/s are identical, shunt motors are preferred because of their effective speed control.

* Speed control can be done from stator side & rotor side (slip ring)

* Stator side →

(1) Supply voltage control

(2) Freq. (V/f)

(3) pole changing

(4) Stator Resistance.

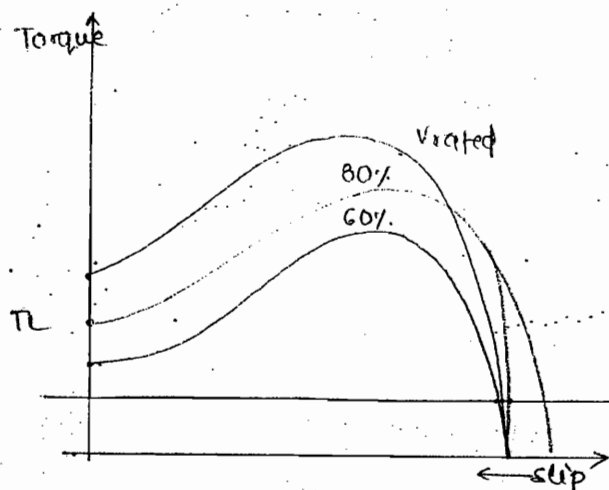
* Rotor side →

(1) Rotor Resistance control

(2) Slip power recovery

(3) Cascading / Tandem connection.

(1) Supply Vol. control →



* It requires an auto Xmer of suitable ratings for getting variable vol. If the motor runs at constant load (or) torque condition as the vol. reduces the motor will react by increasing its slip.

* Consequently speed reduces & operates at different speed.

However for same load the motor will draw more current from supply & becomes inefficient.

②) Freq. (V/f) control $\rightarrow$

- * Freq. is varied by keeping V/f ratio constant in order to maintain flux density constant.
- * Otherwise the stator & rotor core gets deeply saturated & motor draws large magnetising current.
- * It requires expensive power electronic converter ckt for variable vol. & freq.

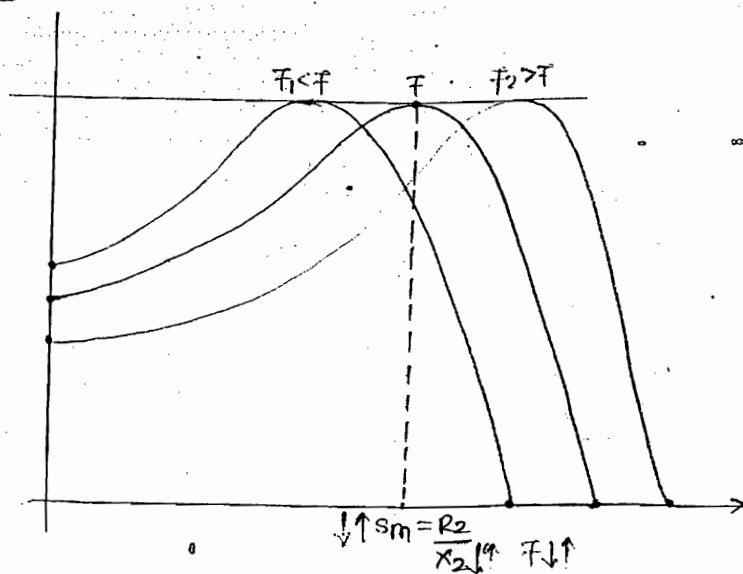
$$V_1 = E = 4.44 f N_1 B_m A$$

$$B_m \propto \frac{V}{f}$$

$$T_{max} \propto \frac{1}{N_s} \frac{E_2^2}{(2X_2)} = \frac{3 \times 60}{2\pi N_s} \left(\frac{E_2^2}{2X_2} \right)$$

$$\begin{array}{c|c|c} f & N_s & f & E_2 & f & X_2 \\ \hline f_1 & N_{s1} = \left(\frac{f_1}{f} \right) N_s & f_1 & E_{21} = \left(\frac{f_1}{f} \right) E_2 & f_1 & X_{21} = \left(\frac{f_1}{f} \right) X_2 \end{array}$$

$$T_{max1} = \frac{1}{N_{s1}} \times \frac{E_{21}^2}{2X_{21}}$$



- * Within the control range the $\max^m$ & breakdown torque remain same.
- * However it doesn't offer wide range as the voltage also correspondingly reduced.
- * Method is very expensive.

(3) Pole exchanging $\rightarrow$

- (1) Multiple wdg.
- (2) Consequent pole changing.

* The stator slot contains 2 independent sets of wdg designed for different no. of pole.

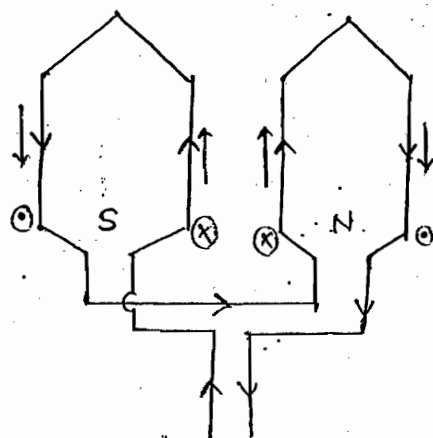
eg:- 2, 4 (OR) 4, 8.

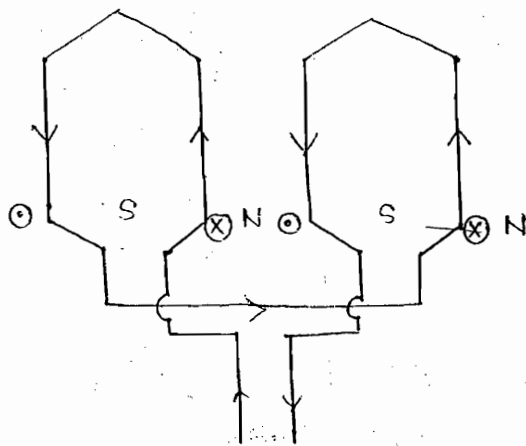
* Depending on the speed requirement either wdg can be connected to the supply & other one left disconnected.

* The wdg are preferably in Δ . This will increase stator size & overall size of motor & motor becomes expensive & economically not advantages for more sets of wdg.

* Another alternate method is through switching the wdg externally for a consequent pole formation in the stator.

* It supports only sq. cage rotor because the poles are formed automatically & the rotor responds to any no. of pole on stator.





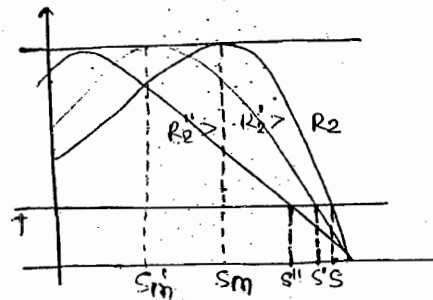
(4) Stator resistance →

- * It is the same method used start the motor.
- * It is advantageous starting than speed control because it will greatly effect η but preferred because of low cost for small rating.
- * Principle is identical to vol. control.
- * It requires only resistances to reduce vol. across stator.

* Rotor side →

(1) Rotor resistance control →

$$T \propto \frac{s}{R_2}$$



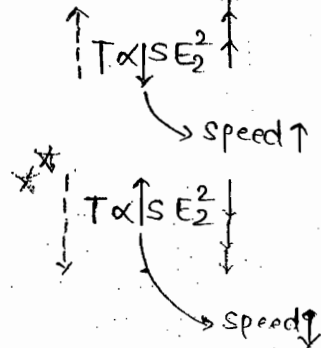
- * In slip ring motors external resistance can be added into the rotor ckt under running condⁿ
- * If the motor runs at const. load (or) torque as the rotor resistance increases slip increase correspondingly to maintain torque constant.
- * As the rotor resistance increased with increased current the rotor loss increases gets overheated practically.
- Therefore it is not suitable for wide range (or) long duration.

(2) slip power recovery →

* Injecting external vol. into the rotor through slip-rings at rotor freq. (or) slip freq. in order to obtain steady torque.

* However this control is also expensive & doesn't give a wide range. It can be done in 2 ways:-

* (1) Injecting the vol. to add the existing rotor vol.



To maintain the torque const. slip reduce & speed increase

* (2) Injecting the vol. to oppose the rotor vol. which reduces the vol. is rotor & to main the torque the motor react by increasing its slip.

consequently speed ↓

'Subsynchronous'

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(3) Cascading/Tandem Connection →

$$N_1 = N s_1 (1 - s_1)$$

$$N_2 = N s_2 (1 - s_2)$$

$$N_1 = \left(\frac{120f}{P_1} \right) (1 - s_1)$$

$$N_2 = \left(\frac{120f}{P_2} \right) (1 - s_2)$$

$$N_1 = N_2 = N$$

$$\left(\frac{120f}{P_1} \right) (1 - s_1) = \left(\frac{120f}{P_2} \right) (1 - s_2)$$

$$\frac{1 - s_1}{P_1} = \frac{s_1 - s_1 s_2}{P_2} \quad (\because s_1 s_2 \rightarrow 0)$$

$$\frac{1 - s_1}{P_1} = \frac{s_1}{P_2}$$

$$\frac{1 - s_1}{s_1} = \frac{P_1}{P_2}$$

$$\frac{1}{s_1} - 1 = \frac{P_1}{P_2}$$

$$\frac{1}{s_1} = \frac{P_1}{P_2} + 1$$

$$\frac{1}{s_1} = \frac{P_1 + P_2}{P_2}$$

$$s_1 = \frac{P_2}{P_1 + P_2}$$

$$N = \frac{120f}{P_1} \left(1 - \frac{P_2}{P_1 + P_2} \right)$$

$N = \frac{120f}{P_1 \pm P_2}$	+ Cumulative cascading. - Differential cascading.
--------------------------------	--

Individual operation:-

$$N_3 = \frac{120f}{P_1}, \quad N_1 = \frac{120f}{P_1 + P_2}, \quad N_4 = \frac{120f}{P_2}, \quad N_2 = \frac{120f}{P_1 - P_2}$$

- * It requires essentially one slip ring motor.
- * Both the motors are mechanically coupled as well as electrically connected to slip rings.
- * This drive will offer 4 sets of speed with independent operation included.
- * If the ϕ seq. of 2nd motor is same it is cumulatively cascaded otherwise differential.

* The poles should be different for 2 motors specially for differential cascading.

(32/29) (C) MFC

Q. → A 50Hz 4p IM is running at 1200 rpm. what should be the freq of vol. injected into rotor for obtaining steady torque?

sol. →

$$s = \frac{1500 - 1200}{1500} = \frac{1}{5}$$

$$sf = \frac{1}{5} \times 50 = 10 \text{ Hz}$$

* Braking →

Braking is done to intentionally to stop motor rapidly at required instant.

This is very essential particularly manufacturing process of drives.

It is similar to braking in dc motors.

(1) Plugging → Reversing the ϕ seq by interchanging any 2 ϕ terminal across stator in running motor.

The rotor want to rotate in the opposite dirn due to which it come near 0 speed where an adequate mech. braking is applied. Otherwise there is a danger of motor rotating in reverse dirn.

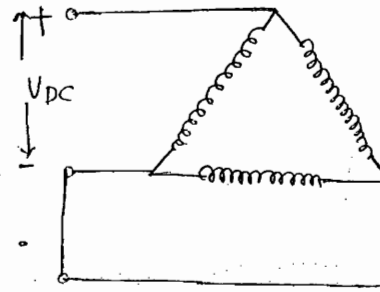
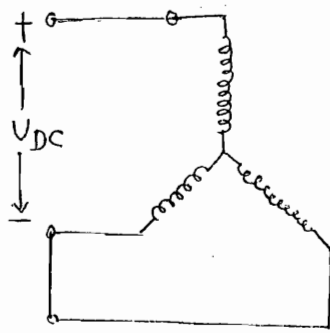
During plugging due to high relative speed ($N_s - N$), high vol. current will appear in rotor. Motors undergo plugging should be design to stand this condn which include high temp. rise.

However the time of plugging is small as the motor starts it will disconnected from supply.

(2) Dynamic → Disconnecting 3 ϕ supply across stator & a suitable dc vol. will be applied across any 2 phases. it produces a constant stationary time in varying flux in air gap & make rotor to stop quickly.

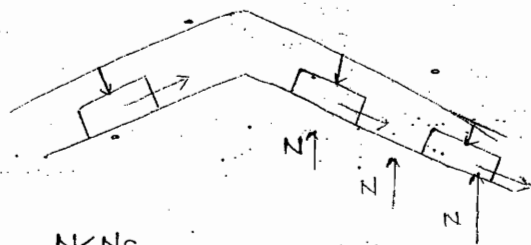
As the rotor rotates due to flux, emf, current & torque produce to oppose cause (relative speed)

Therefore rotor stops quickly.



Note → There is no danger of speed reversal but induced currents in rotor should be control either from stator side or rotor side if its a slip ring.

(3.) Regenerative Braking →



$$N < N_s$$

$$s: 1-0$$

$$N > N_s$$

$$-s = \frac{N_s - N}{N_s}$$

$$-E_2 -I_2 -T$$

$$I_1 \begin{cases} I_0 \\ -KI_2' \end{cases}$$

- * This is not intentional braking & it won't stop the motor like above 2 b.
- * But it will naturally produced a braking torque when motor is subjected to overhauling load & release some stresses for mech. braking sys.
- Therefore it is advantageous in mountain railways where slip ring motors are preferred.
- * When the motor increases its speed ($N > N_s$) slip becomes -ve to develop -ve torque which produce braking.
- * The rotor induced emf & current also reverses & consequently rotor component in stator will reverse.
- Therefore during this braking period it can act as Ind'n genr while supplying the rotor component of current to supply lines instead of drawing it.
- * In the period it continues to draw magnetising current from supply.